

Business Sketch

TRACE (Triage, Report, Action, Capture, Escalate) is an open-source AI platform designed to assist field service technicians with safe, efficient, and auditable troubleshooting workflows. It demonstrates how multi-agent AI, combined with synthetic datasets, can guide junior technicians while maintaining human oversight and traceability.

TRACE provides an Escalation Agent that requests human approval for unsafe or critical tasks from the organizations that manage engines, such as engineers, mechanics, or maintenance teams. It also records all AI decisions with inputs, outputs, and timestamps. It stores contexts locally and syncs once online. The system analyzes engine performance data, identifies potential problems before they occur, and provides recommendations for the necessary actions. This would help save time and retain valuable information, unlike other AI agents, which help us answer questions but don't assist with the tasks we need.

This Agentic AI would be used in Organizations that have field service technicians. This would serve as a guide for a junior field Technician using AI (Artificial Intelligence). It shows them the patterns of risks that could occur or have occurred, and helps mitigate and prevent potential threats.

The KEY activities that TRACE AI performs include

The Engine Data: TRACE collection gathers information on engine performance, such as speed (RPM), fuel usage, and temperature. This information comes from sensors or from technician input.

Recommendations: after the analysis, TRACE informs the team of the risks and includes maintenance before the threats occur. It also keeps records of past occurrences to track similar threats and future risks, and notifies them as soon as possible.

The Cost of TRACE AI would be spread across the first and second halves of the year for software maintenance and upgrades to ensure the AI Agents perform their tasks accurately and appropriately. This would help TRACE AI keep track of its agents and ensure the organization's data is secure and the information provided to the AI Agent is safe.

TRACE AI would be monetized through a \$5,000 monthly subscription. We came to this conclusion because the monthly subscription covers high-frequency API processing, secure cloud storage for engine telemetry, and continuous fine-tuning to make sure the agents remain accurate as the engine technology evolves. While no system is impenetrable, TRACE uses human-in-loop triggers to ensure that no critical action is taken without authorization. This AI doesn't only work FOR you, it works WITH you!